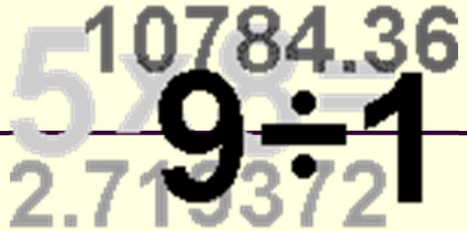
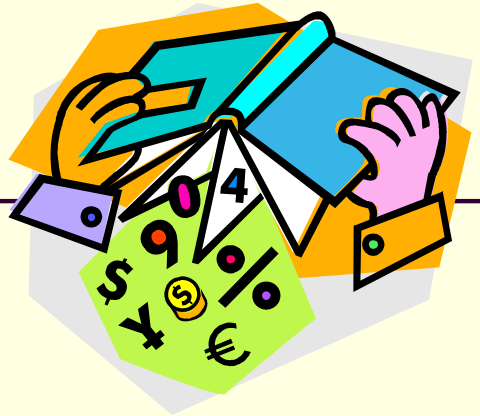
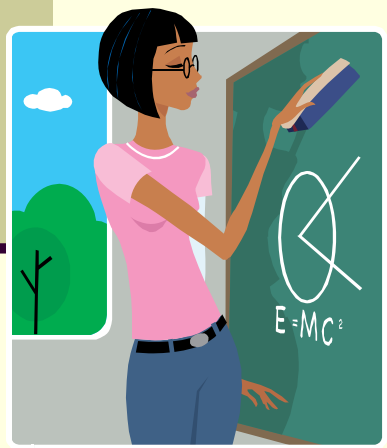




Teaching Approaches and Learning Theories (Part 1)



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My Teaching Philosophy

- As a maths teacher, you don't teach maths. You teach …?
- You teach _____. What is the difference?
- Teaching _____ means:
  - you **care** whether they learn maths or not, e.g. you don't just deliver lessons to cover syllabus
  - you **care** whether they learn other subjects
  - you **care** about **their entire well-being**, not just their academic performance

My Teaching Philosophy

- **Teaching ≠ Learning**
- Our duty is not to teach *per se*, but to teach so that _____ takes place
- Unfortunately, George Bernard Shaw said, "He who can, does. He who cannot, teaches."
- Many people **wrongly** equate teaching with talking

My Teaching Philosophy

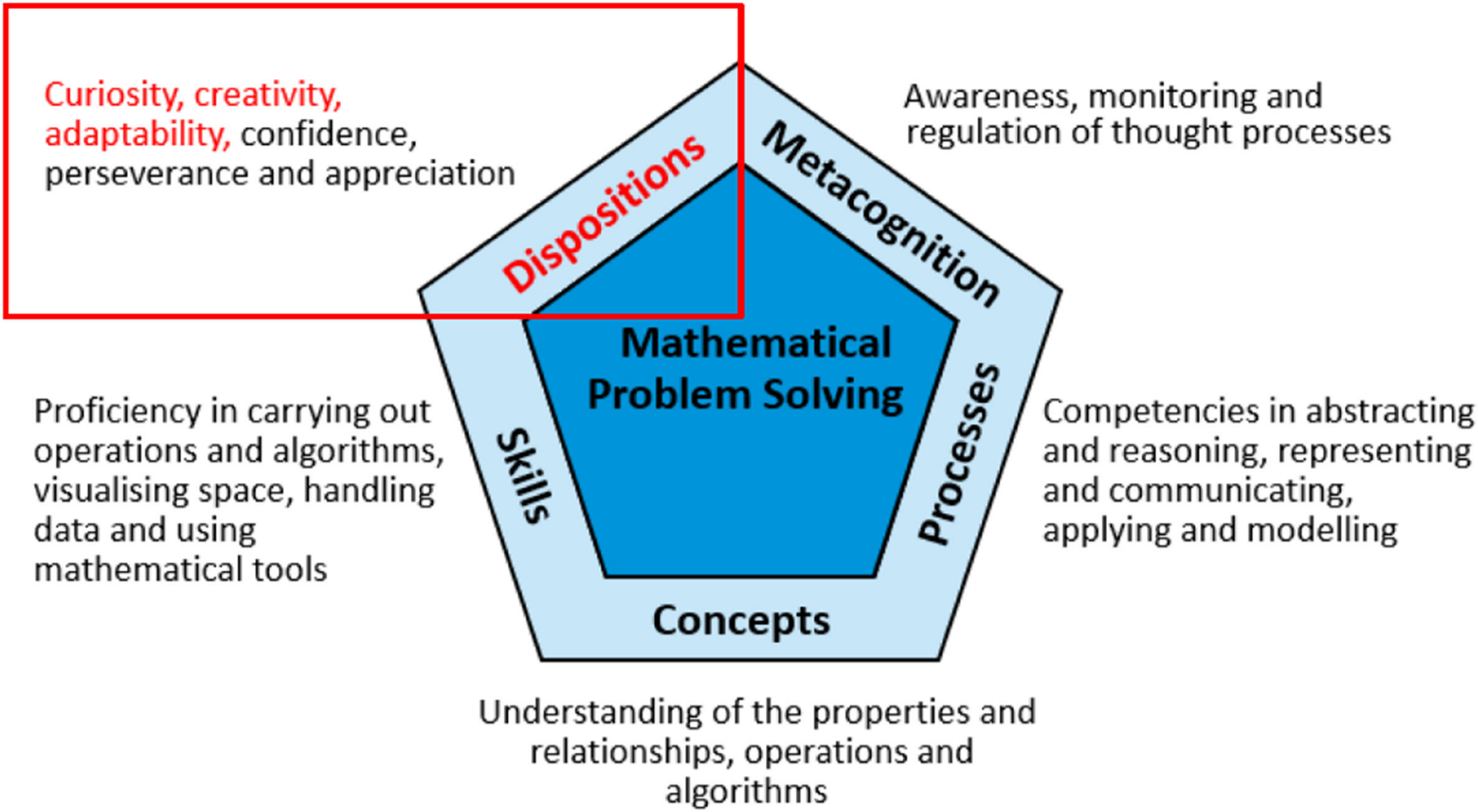
- "It's easier said than done." "Talk only."
- Talking is easier than doing
- But _____
  (as comic has shown): teaching such that the other person **learns** is often more difficult than doing it yourself!
- Talking is the easiest, followed by doing; hardest is teaching others **to do**

Key Lesson 1

- As a maths teacher, you don’t teach maths.

  You teach _____.

Mathematics Curriculum Framework



Relational vs. Instrumental Understanding

- Skemp[2] (1976) distinguished between two types of understanding:
  - _____ **understanding**: knowing both what to do and **why**
  - Instrumental understanding: rules without reasons
- Is 'instrumental understanding' a type of understanding?

[2] Skemp, R. R. (1976). Instrumental understanding and relational understanding. *Mathematics Teaching, 77*, 20-26.

Relational vs. Instrumental Understanding

- Skemp (1976) regarded **‘instrumental understanding’** as _____ **at all**
- Until he realised many teachers and students believe that ‘instrumental understanding’ is understanding
- E.g. if students know how to solve quadratic equation by equating both factors to 0, then they have **understood** how to solve quad. eqns**?!**
- But why equate both factors to 0?

Relational vs. Instrumental Understanding

- To many teachers and students, not important to understand why equate both factors to 0
- In other words, instrumental understanding (i.e. no understanding) is **the only understanding** that they know, and they don't want to teach or learn relational understanding
- So _____ do many students still have so much difficulty if they just learn skills and rules without (relational) understanding?

Know Your Students' Abilities

- You may say, "But I have no trouble understanding secondary school maths!"
- You were able to study in a university while many students can't even study in a polytechnic
- Some **failed O-level maths so badly** their F9 was not just below 40% but can be like _____
- Some trainees who came back from Teaching Practice said they were shocked some **Sec 3** students could not even evaluate 3 – (–2), or use a protractor to measure an angle

Know Your Students' Abilities

- You could memorise mathematical rules without understanding in secondary school, but **when did you reach the stage when you _____ having difficulty?** JC? University?
- If you didn't understand a theorem or its proof in university, was it easy to remember?
- After you had understood it, was it **easier** to remember? (easier ≠ easy)
- It does not mean you will never forget

Know Your Students' Abilities

- When you forgot, was it **easier** to pick it up again during revision if you had _____ it before?
- In conclusion, was it **harder to remember** (or recall when you forgot) if you **didn't understand** (or understood it before)?

Key Lesson 2

- Many students, whom you will be teaching, have **reached the stage when they start having _____ in maths in secondary school** (or even primary school)
  - Teaching rules without understanding is **not** going to help them: they can't cope with unrelated rules anymore

Constructivism

- But **why** is it that many students cannot learn rules without understanding?
- Socrates: "Education is the kindling of a flame, not the filling of a vessel."
- Students are **not empty vessels** for teachers to pour or transmit knowledge. Why?
- **Our brain works** by linking new knowledge to existing knowledge _____
- **Weak links (or bonds)** are easily broken, so students will not remember the new knowledge

Constructivism

- Called **theory of** _____: students learn by constructing their own knowledge
- The term 'constructivism' can be traced further back to **Jean Piaget** (1896 – 1980; 84 years old)
- **Schema**[3] is a mental framework that is created as people interact with their physical and social environments; **structured cluster of concepts**; used to represent objects, scenarios or sequences of events or relations

[3] Original idea of schema (plural: schemata) from Immanuel Kant

Constructivism

- Piaget: two processes to create a schema to perceive and / or interpret their experiences
  - _____: new knowledge is assimilated **without changing** existing knowledge
  - _____: existing knowledge needs to be **changed** to accommodate new knowledge
- Basis for theory of constructivism: we learn by **constructing our own knowledge meaningfully**, linking it to existing knowledge

Learning Theories

- Most people don't like theories: 'talk only'
- Why must people go for driving theory test?
- Passing theory test does not mean you can drive
- Suppose you **only** passed practical test
- While driving, you see this sign
- What does this sign tell you?
- It may **cost you your money, your convenience or even your _____!**

Learning Theories

- **Theories can inform practice**
- Need to distinguish two groups of theories:
  - Untested theories: "This is what I think", but it may not work in practice
  - Theories that are **well _____ in research**
- Theories in education or social sciences cannot be proven by research as rigorously as scientific laws (conditions in scientific experiments can be repeated to verify the laws)

Learning Theories

- Education theories have too many _____ that cannot be repeated (e.g. different types of pupils, different pupils of same type, different teachers, different classroom environments, etc.)
- Does it mean that learning theories grounded in research are also not useful?
- E.g. research has found that some (but not all) students have certain misconceptions
- So should we ignore it?

Learning Theories

- _____, we should try to ensure that these students in our class (whoever they are) should not develop these misconceptions
- In Teaching Scenario 1, if the teacher has paid attention to all these learning theories, he would not have **wasted one entire month** trying to teach by talking without seatwork supervision
- To summarise, learning theories **grounded in research** are _____ **for you**

Pentagon Model: Concepts

- Rules without understanding will confuse low readiness or even middle readiness students, so the need for **concept development** (to teach for conceptual or relational understanding)
- So how to **teach for understanding**?

How to Teach Prime Numbers

- Do you start with definition? Why or why not?
  - Most students will _____ understand definition (recall your university maths?)
  - Motivated students will wait for teacher to explain by giving examples
  - Unmotivated students will _____ because it just confirms that they don't know maths
  - Even in a good class, there are low readiness students in maths

How to Teach Prime Numbers

- Do you start with definition? Why or why not?
  - Most students will _____ understand definition (Skemp, 1987)
  - Recall your university maths?
  - Motivated students will wait for teacher to explain by giving examples; unmotivated students will _____ because it just confirms that they don't know maths (even in a good class, there are low readiness students in maths)
  - So how to teach?

Principle of learning of maths[4]

(1) Concepts of a higher order than those which people already have cannot be communicated to them by a definition, but only by arranging for them to encounter a suitable collection of examples.

The first of these principles is broken by the vast majority of textbooks, past and present. Nearly everywhere we see new topics introduced not by examples but by definitions, definitions of the most admirable brevity and exactitude for the teacher (who already has the concepts to which they refer) but unintelligible to the student. For reasons which will be apparent, examples cannot be quoted here, but readers are invited to verify this statement for themselves. It is also a useful exercise to look at some definitions of ideas new to oneself in books about mathematics beyond the stage which one has reached. This enables one to experience at first hand the bafflement of the younger learner.

Good teachers intuitively help out a definition with examples. To choose a

[4] Skemp, R. R. (1987). *The psychology of learning mathematics*. Hillsdale, NJ: Erlbaum.

How to Teach Prime Numbers

- You can start by building upon their _____ _____ (recall learning theory of **linking** new knowledge to existing knowledge): ask students how to find all the factors of 18
- Write on the whiteboard: $18 = 1 \times 18$

$$= 2 \times 9$$

$$= 3 \times 6$$

$\therefore$ factors of 18 are 1, 2, 3, 6, 9 and 18

How to Teach Prime Numbers

- Can get students to tell you factors of other numbers, but you engage very few students
- Another way is to give students **worksheet** to find factors of 1 to 20, so that ____ **students do**
- Still need to start worksheet by asking and listing factors of 18 on whiteboard so that every student knows how to fill table
- Discuss with the person beside you your answer to **Q2** in worksheet

How to Teach Prime Numbers

- If teaching low readiness students, you can give **verbal hints** on what meaningful ways in Q2
- Or you can provide more **scaffolding** from the start by rephrasing Q2: divide 1-20 into three groups according to the number of factors
- Group 1: Nos. with only ____ factor
- Group 2: Nos. with only ____ different factors
- Group 3: Nos. with more than 2 different factors (why not have more groups?)

How to Teach Prime Numbers

- Nos. with only 3 diff. factors, nos. with only 4 diff. factors, etc. are not very different among themselves (or else infinite no. of groups), but they are diff. from special type of nos. in Gp 2
- Let us give each group a name
  - Numbers in Group 2 are called prime numbers
  - Numbers in Group 3 are called composite nos.
  - Group 1 has no name: contains the number 1
- **Guide students to** _____ prime numbers

How to Teach Prime Numbers

- Now can you ask, “What is the definition of prime numbers?”
- Some students are still frightened by the word ‘definition’, so they can’t answer your question
- Instead, ask, “So, _____ a prime number?”
- Definition: A prime number is a whole number that has **exactly** two **different** factors
- Students will only learn ‘integers’ later, so cannot use ‘positive integer’ in definition yet

How to Teach Prime Numbers

- MOE
  (LE) N1(a): Classify whole numbers based on their number of factors … (O-level Maths Syllabus 2013, 2020)
- **MOE wants students to go through all these LE**, so MOE-approved textbooks will contain all the LE for teachers to teach

How to Teach Prime Numbers

- Teaching Scenario 1: Teacher asked whether they understood and whole class answered ‘yes’
- Does _____ mean they really understood. Why?
  - Not all students answered ‘yes’ (it may just be a few students with a very loud voice)
  - May feel ashamed to say they don’t understand
  - You may ask those who said ‘no’ what they don’t understand, so better for them to say ‘yes’
- Need to **ask individual students questions that test for understanding**

Q3

- by asking: “If a whole number is not prime, then it must be composite. Do you agree? Explain.” **Please write down your answer to this question below.**

Pentagon Model: Concepts

- Key idea for teaching for understanding: make students **think** so that they can construct the knowledge **meaningfully**
- One method to do so is **Bruner's** _____ _____ **learning**: guide students to think and discover concept
- Can be **teacher-centred** (teacher stands in front and guides whole class) **or student-centred** (students do guided-discovery worksheet)

Key Lesson 3

- Guide students to _____ and **discover** mathematical concepts so that they can **construct** their own knowledge **meaningfully**

Pentagon Model: Processes

- Guided-discovery learning make students think: overlap of (developing) **concepts** and (thinking) **processes in Pentagon Model**
- To develop process, you can **teach students** _____ when solving problems or going through worked examples, e.g. how to prove trigonometric identities
- Prove this identity:

$$\frac{2\cos x + \sin 2x}{2\cos x - \sin 2x} = \tan^2 x + 2\tan x \sec x + \sec^2 x$$

Proving Trigonometric Identities

- **Strategy 1: Start from the more complicated expression** (because it is easier to simplify)
- **Strategy 2:** Think of what _____ you can use at each step
- **Strategy 3: Work from** _____ if you are stuck halfway
- But you **should present the working** from LHS to RHS, or from RHS to LHS

Proving Trigonometric Identities

- Students who are good at proving identities are those who can remember the basic identities
- If not, how to apply Strategy 2 (think of what basic identities you can use at each step)
- **Need to memorise basic identities** (after understanding how they are derived)
- Basic knowledge becomes useful needed to solve problems (Schoenfeld[5], 1985)
- **Thinking does not exist in a vacuum**

[5] Schoenfeld, A. H. (1985). *Mathematical problem solving.* Orlando, FL: Academic Press.

Key Lesson 4

- **Teach students how to _____** when solving problems or going through worked examples

Pentagon Model: Metacognition

- **Metacognition: _____ about thinking**
  - Be **aware** of your own thought processes
  - **Monitor your own thinking**, e.g. during problem solving
  - **Self-regulation of learning**: as a trainee teacher, you need to take charge of your own learning; can teach students to do the same as well

Pentagon Model: Metacognition

- One way to **self-regulate learning** is to do _____ after every lesson
- Research has shown that reflection after every lesson has led to significant improvement in test performance
- How many takeaways for every lesson?
- If students can remember one takeaway for a long time, it is good enough
- In-service course on metacognition

Summary of Learning Theories

- Theory of **constructivism**
- Piaget's **assimilation** and **accommodation** processes to create schema
- Skemp's instrumental understanding and **relational understanding**
- **Bruner's guided-discovery learning**

Conclusion for this Lecture

- Ask yourself "What is _____ teaching philosophy?"
- Unrelated meaningless rules are not helpful: need to **teach for conceptual understanding**
- Need to teach students **how to think** when solving problems or going through worked examples
- Need to get students to **reflect** at end of lesson

Next Lecture

- One more method to develop math concepts
- How to develop procedural skills
- How to develop positive attitudes

Reflection

- Write down **‘most important’ takeaway** for this lecture that you would like to **remember for a long time** to come